Repayment Resilience | Product Case Study

Independent portfolio project | Tina Liang | November 2026

Synthetic data only. No KOHO data or deployed product; measured effects are constructed for a reproducible analytical demonstration.

Problem & product hypothesis

Fixed repayment dates can miss customers' income cycles. Offer an opt-in pay-date-aligned due date; pair this with human-reviewed hardship assistance to improve repayments without worsening customer harm.

MVP and Phase 2

Phase 1: eligibility, due-date preview, consent, audit trail, repayment events, manual hardship intake and rollback. Phase 2: design hardship or settlement lending with affordability reviews, underwriting, clear terms and Credit Risk, Finance, Legal, Compliance, partner and Engineering sign-off.

Experiment and feasibility

Randomize by customer, 50/50; primary: next-payment success (intention-to-treat). Track 30/60/90 DPD, NSF, complaints, hardship and redefault as guardrails.

Power, 72% baseline; two-sided alpha 5%; 80% power: 2 pp MDE = 7,733/arm; 3 pp = 3,395/arm; 5 pp = 1,191/arm. Current synthetic experiment has 3,408 customers total, insufficient for a 3 pp test.

Simulation outputs (not forecasts)

6,000 synthetic accounts; 3,408 in randomized demonstration. Next-payment success: aligned 80.6% versus control 72.0%, observed difference 8.62 pp (95% CI 5.78 to 11.47 pp). A 7.5 pp effect was injected into the generator. 179 accounts flagged for manual hardship review.

Settlement and on-load collections

Settlement decision: expected net recovery with the offer must exceed expected net recovery under ordinary collections, after costs, and must pass affordability and fair-treatment review. On-load collections could improve recoveries but may deplete essential spending funds: require opt-in, caps, exclusions and complaint/hardship guardrails.

Economics and go/no-go

Avoided losses = (control charge-off roll rate - treatment charge-off roll rate) x balance at risk x loss rate at charge-off. Do not calculate savings without observed charge-offs and recovery data. Scale only if evidence is credible, operational safeguards pass, and legal/risk approvals are secured.

Evidence and next step

Code: src/analysis.py and src/power_analysis.py. SQL metric template, PRD and synthetic dataset included. Validate actual customer eligibility, payment rails, repayment cadence, cohort maturities and production economics before any real pilot.

Independent product analysis, not affiliated with KOHO.